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Python Test

oops

1) what is object-oriented programming (oop)
→ Object-oriented programming paradigm means 'way of thinking'. ✓ ①

2) what are the four pillars of oop? Explain
→ There are four pillars of oop.

1] Inheritance :- ✓

2] Polymorphism :- polymorphism means "Many Forms". ✓ ①

3] Encapsulation :- Encapsulation is data hiding ✓

4] Data Abstraction :- Data Abstraction means hiding implementation details. ✓

3] Explain difference between class & object with example
→ class object

class	object
① class is a blueprint is used to create object.	① an object is instance of a class

② eg: class student:
 def display(self):
 print("welcome
 to the class")

s1 = student()
s1.display()

② eg:- ✓
s1 = student()

4) what is inheritance? Explain its types.

→ Inheritance :- one inheritance allows child class and parent class one class is inherits from another class. ✓ ①

Types of Inheritance

- ✓ a) Single Inheritance: child class inherits from only one parent class.
- ✓ b) Multiple Inheritance: child class inherits from more than one parent class.
- ✓ c) Multilevel Inheritance: child inherits from parent & parent inherits from grand parents.
- ✓ d) Hierarchical Inheritance: many child classes inherit from one parent class.
- ✓ e) Hybrid inheritance: It is a combination of one or more inheritance types.

5) What is polymorphism? Explain method overriding & method overloading?

→ Polymorphism means "many forms".

1) Method overriding: Same method and same parameters.

2) Method overloading: Same methods but different parameters.

6) What is Encapsulation? its advantage.

→ Encapsulation means data hiding and protecting data.

Advantage :- 1) Data is protected 2) security

7) What is Abstraction? How it is implemented in python?

→ Abstraction shows what an object does, not how it does it.

8) Difference betⁿ Abstraction & Encapsulation

Encapsulation	Abstraction
① It is focus on data hiding	① It is focus on implementation
② Achieved by Access modifiers	② Achieved by Abstract class

1] Write a program to print the multiplication table of a given number using for loop.

→ `num = int(input("Enter a number:"))`
`for i in range(1, 11):`
`print(num, "*", i, "=", num * i)`

o/p :- Enter a number : 2

$$2 * 1 = 2$$

$$2 * 6 = 12$$

$$2 * 2 = 4$$

$$2 * 7 = 14$$

$$2 * 3 = 6$$

$$2 * 8 = 16$$

$$2 * 4 = 8$$

$$2 * 9 = 18$$

$$2 * 5 = 10$$

$$2 * 10 = 20$$

2] write a program to print all prime number between 1 and 100 using a for loop.

→ `for num in range(1, 100):`
`prime = True`
`for i in range(2, num):`
`if num % i == 0:`

`prime = False`

`if prime: break`
`print(num)`

① write a program to check whether a given number is palindrome using while loop
→ num = int(input("Enter a number :"))
while num > 0:

② write a program to reverse a given number using while loop.

a = 10

reverse = 0

while a > 0:

reverse = reverse * 10 + a

print("Reverse" =', reverse)